

EDA - QUANTITATIVE AND QUALITATIVE ANALYSIS

AIM

To understand the importance of Exploratory Data Analysis (EDA) by performing quantitative and qualitative analysis on the 'tips' data set.

PROGRAM

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

tips = sns.load_dataset('tips')
print(tips.head())

# ----- Quantitative analysis -----
print(tips.describe())

sns.displot(tips.total_bill, kde=True)
plt.show()

sns.displot(tips.total_bill, kde=False)
plt.show()

sns.jointplot(x=tips.tip, y=tips.total_bill)
plt.show()

sns.jointplot(x=tips.tip, y=tips.total_bill, kind="reg")
plt.show()

sns.jointplot(x=tips.tip, y=tips.total_bill, kind="hex")
plt.show()

sns.pairplot(tips, height=2)
plt.show()

print(tips.time.value_counts())
sns.pairplot(tips, hue='time', height=2)
plt.show()

sns.pairplot(tips, hue='day', height=2)
plt.show()

sns.heatmap(tips.corr(numeric_only=True), annot=True)
plt.show()

sns.boxplot(y=tips.total_bill)
plt.show()

sns.boxplot(y=tips.tip)
plt.show()

# ----- Qualitative analysis -----
print(tips.day.value_counts())
print(tips.sex.value_counts())

sns.countplot(x='day', data=tips)
plt.show()
```

```
sns.countplot(x='sex', data=tips)
plt.show()

tips.sex.value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.show()

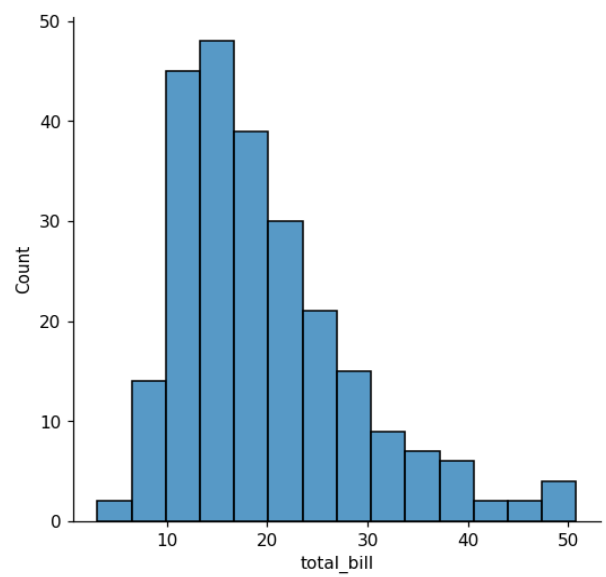
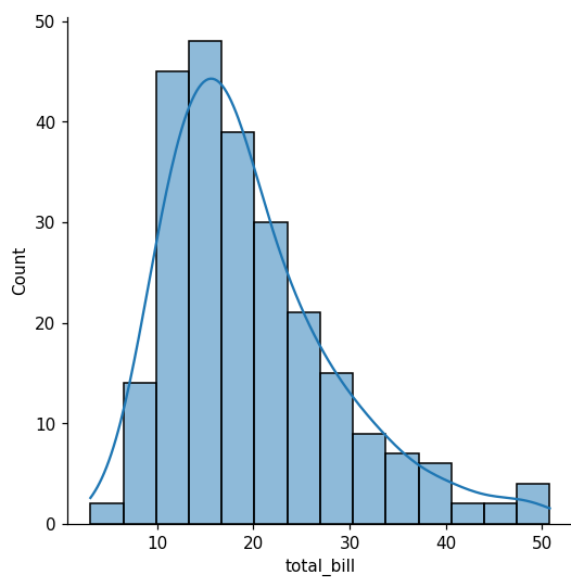
tips.sex.value_counts().plot(kind='bar')
plt.show()

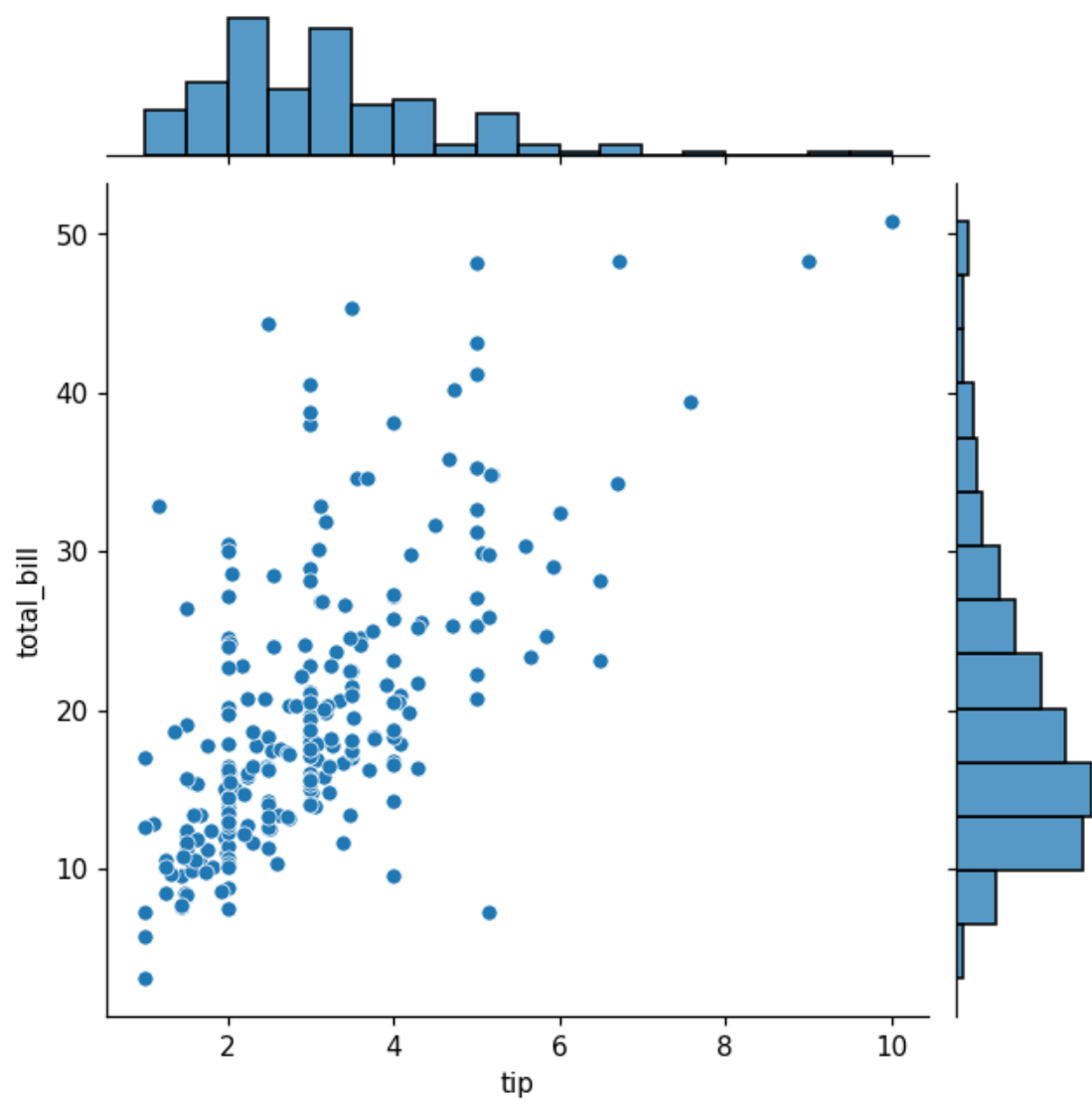
sns.countplot(x='day', data=tips[tips.time == 'Dinner'])
plt.title('Dinner count per day')
plt.show()
```

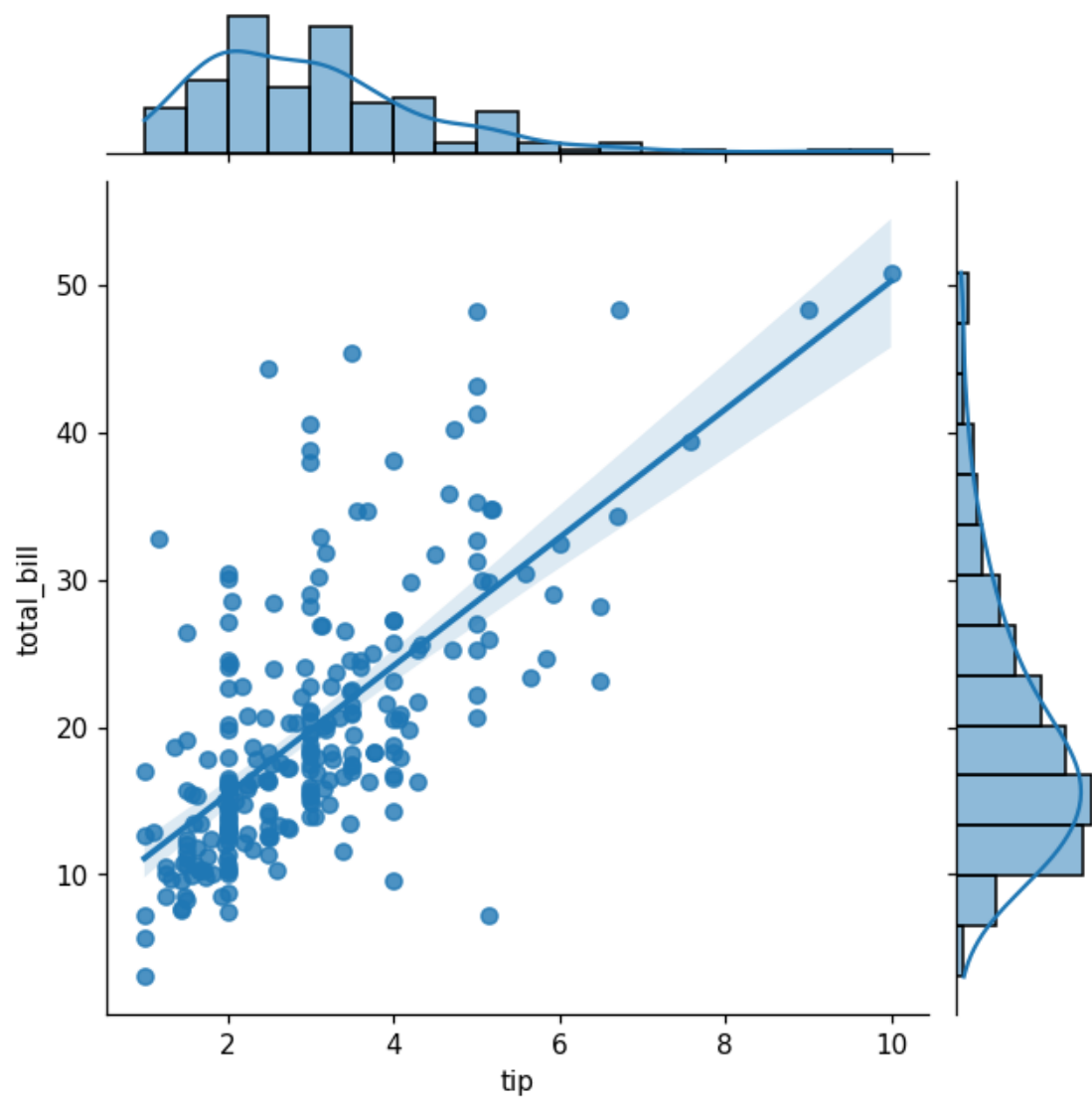
OUTPUT

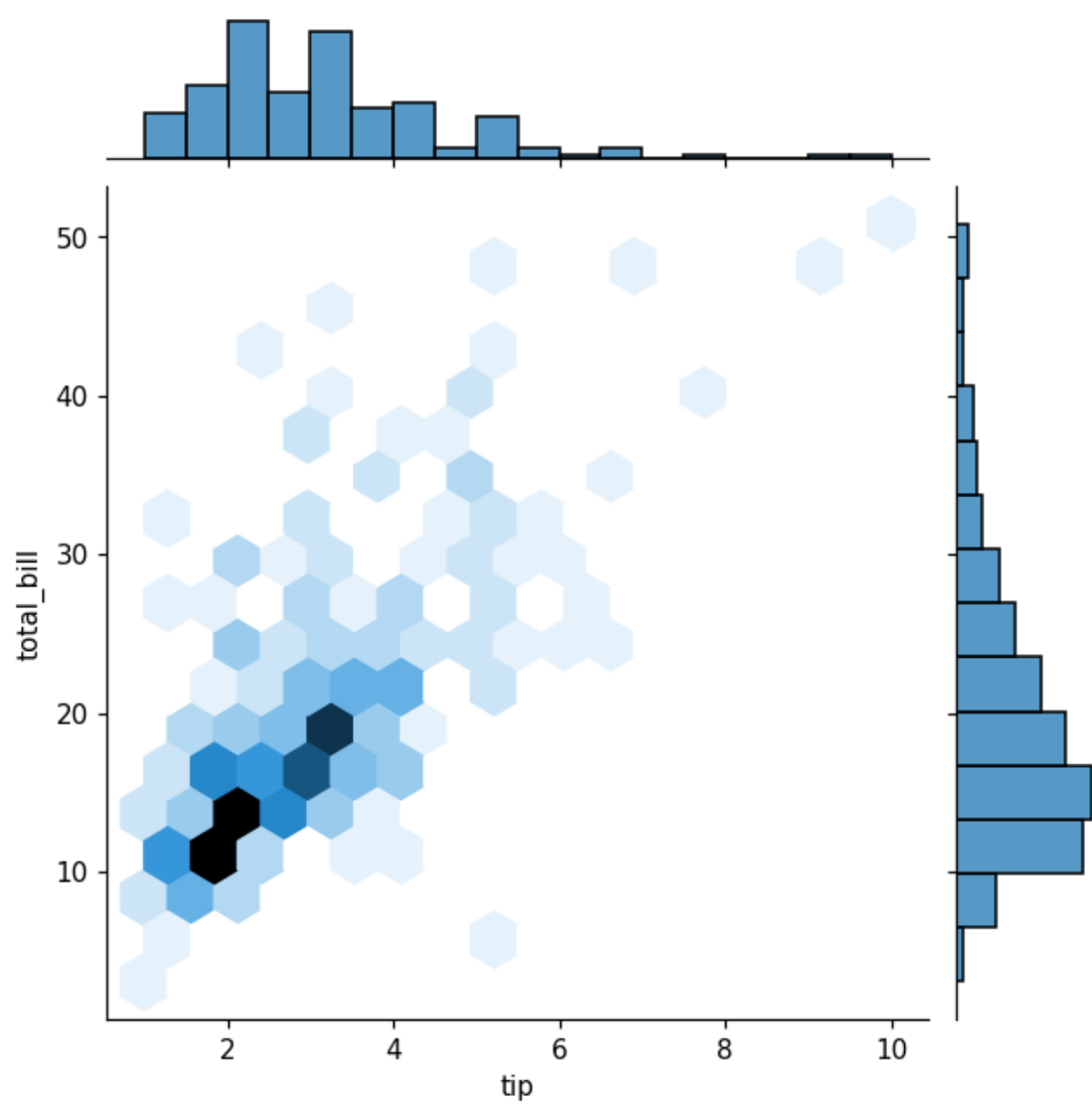
	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

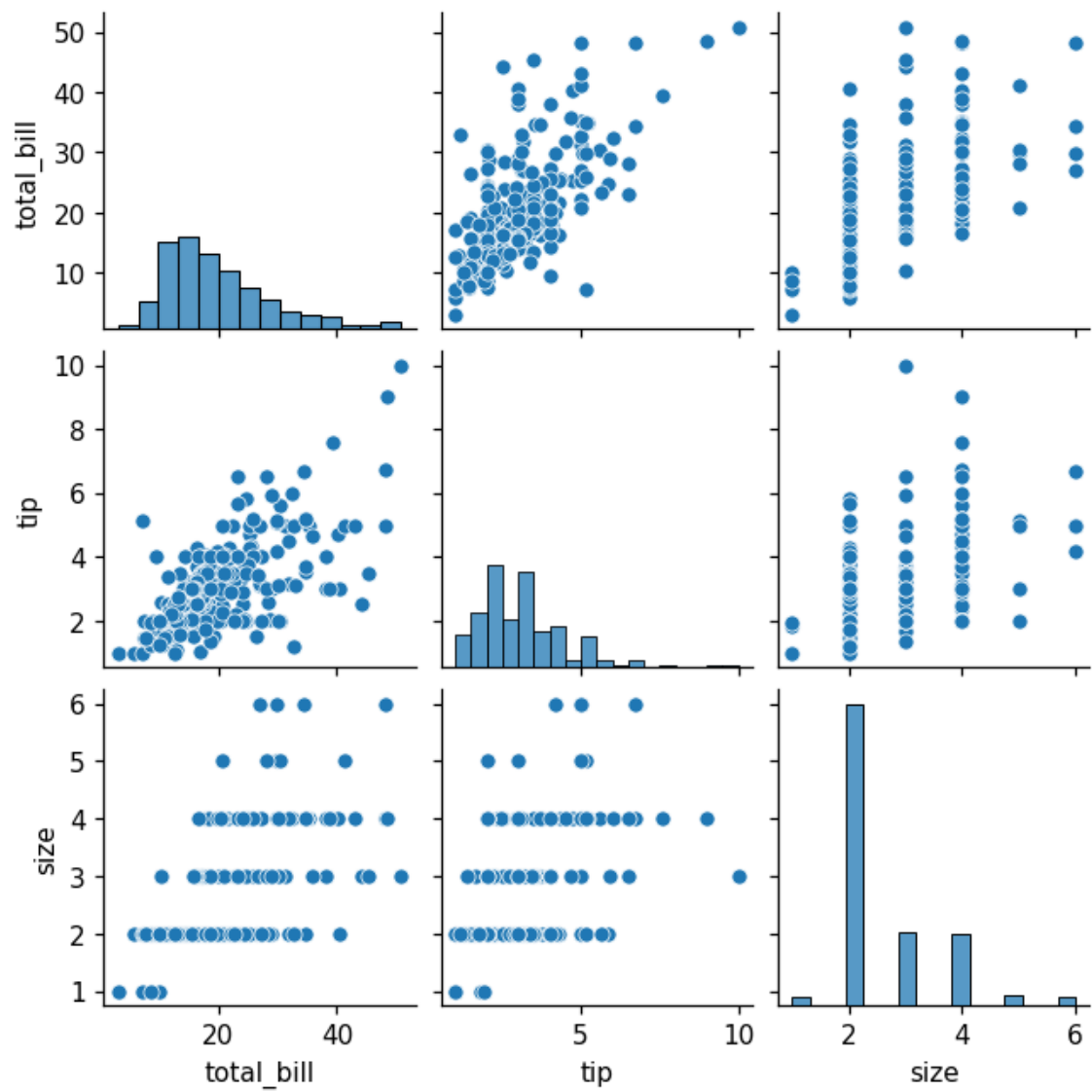
	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000



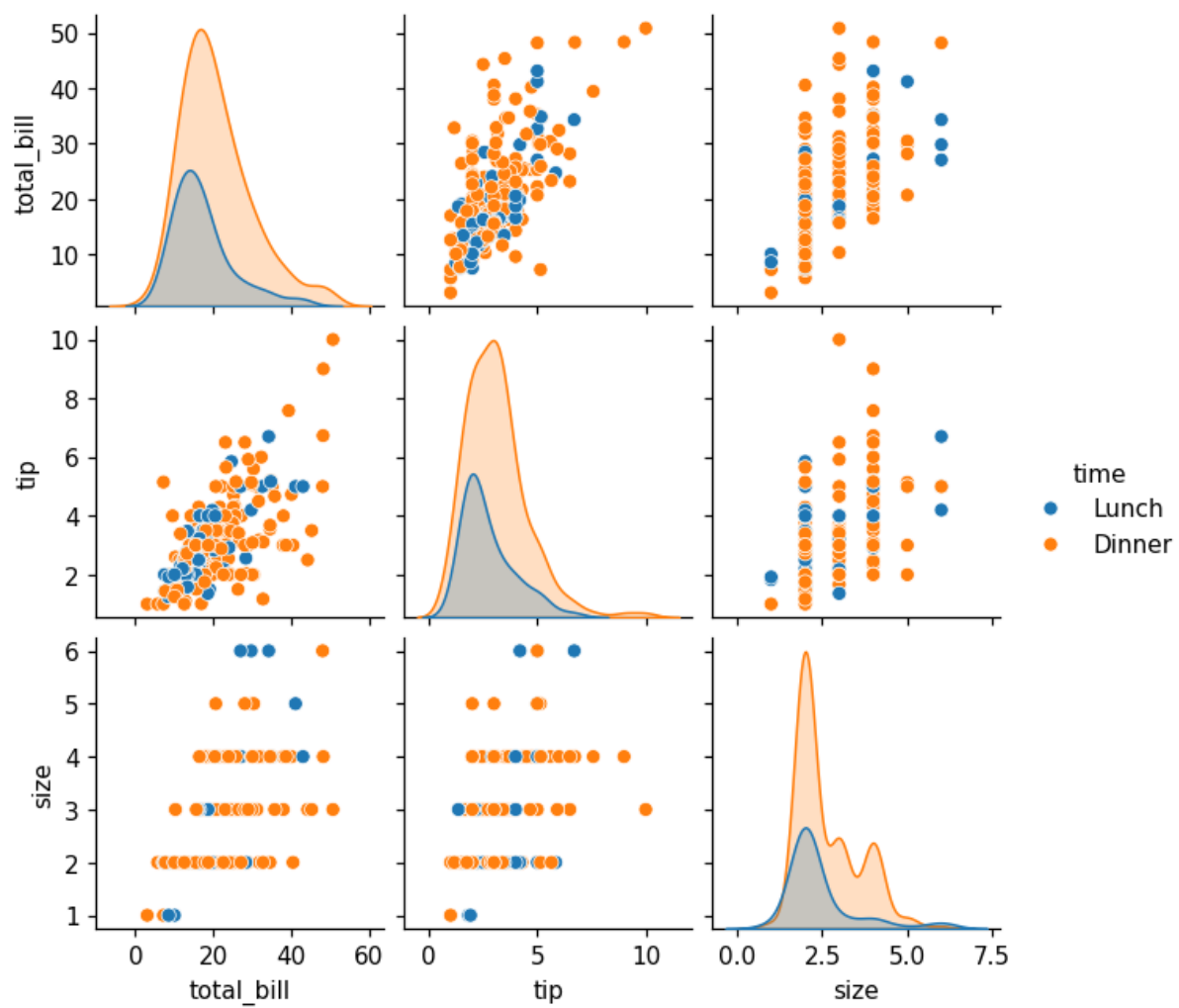


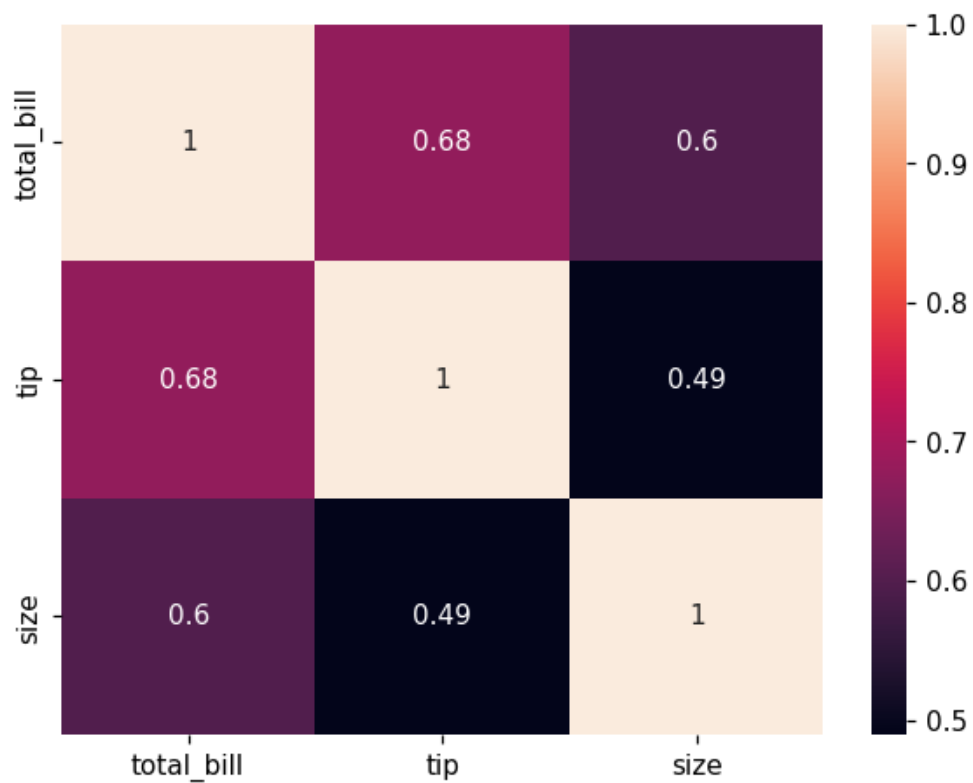
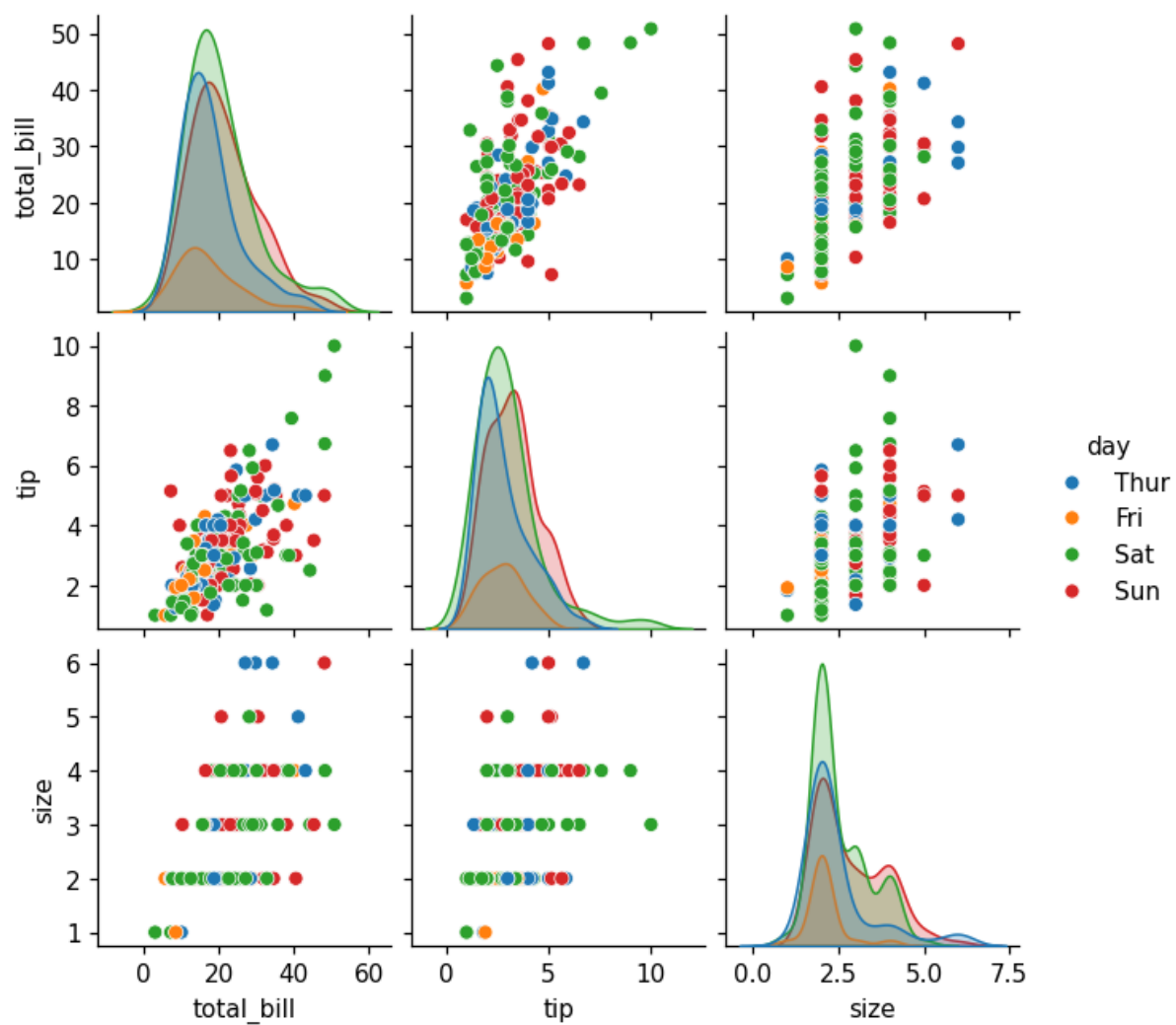


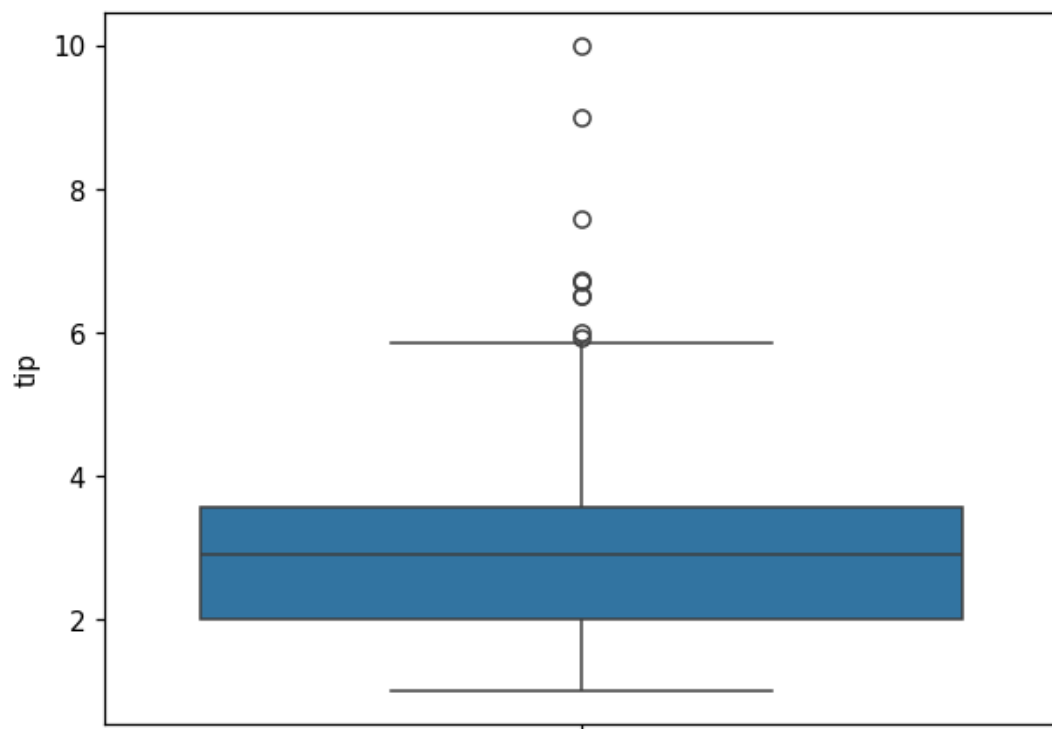
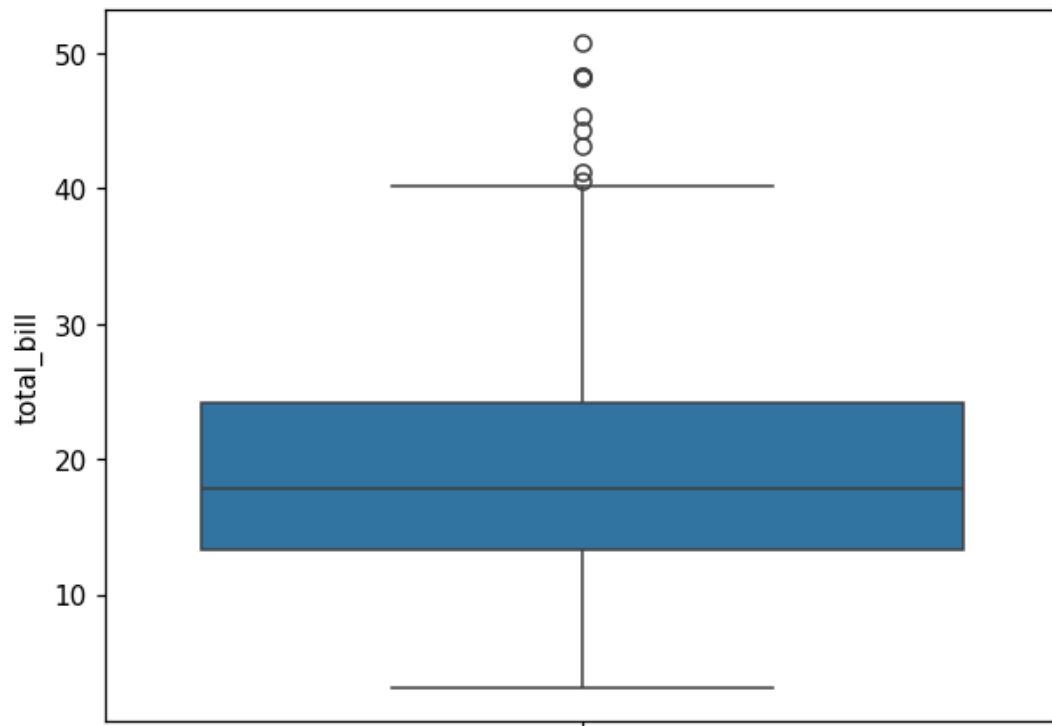




```
time
Dinner    176
Lunch      68
Name: count, dtype: int64
```



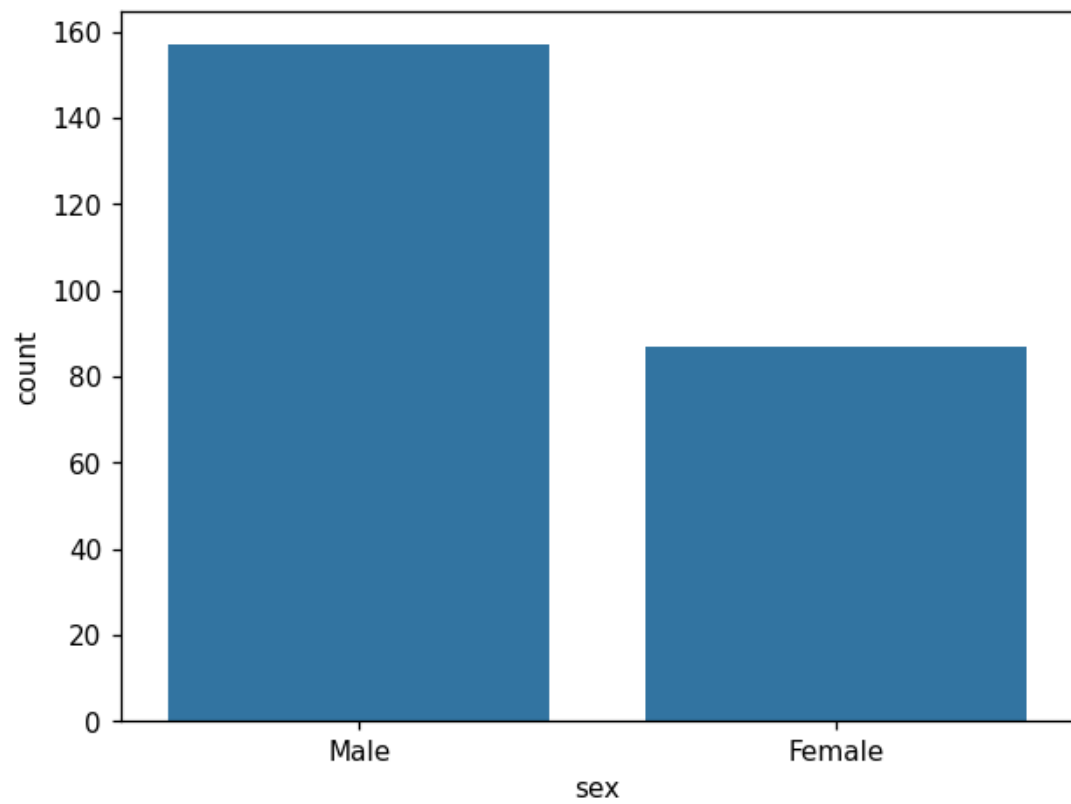
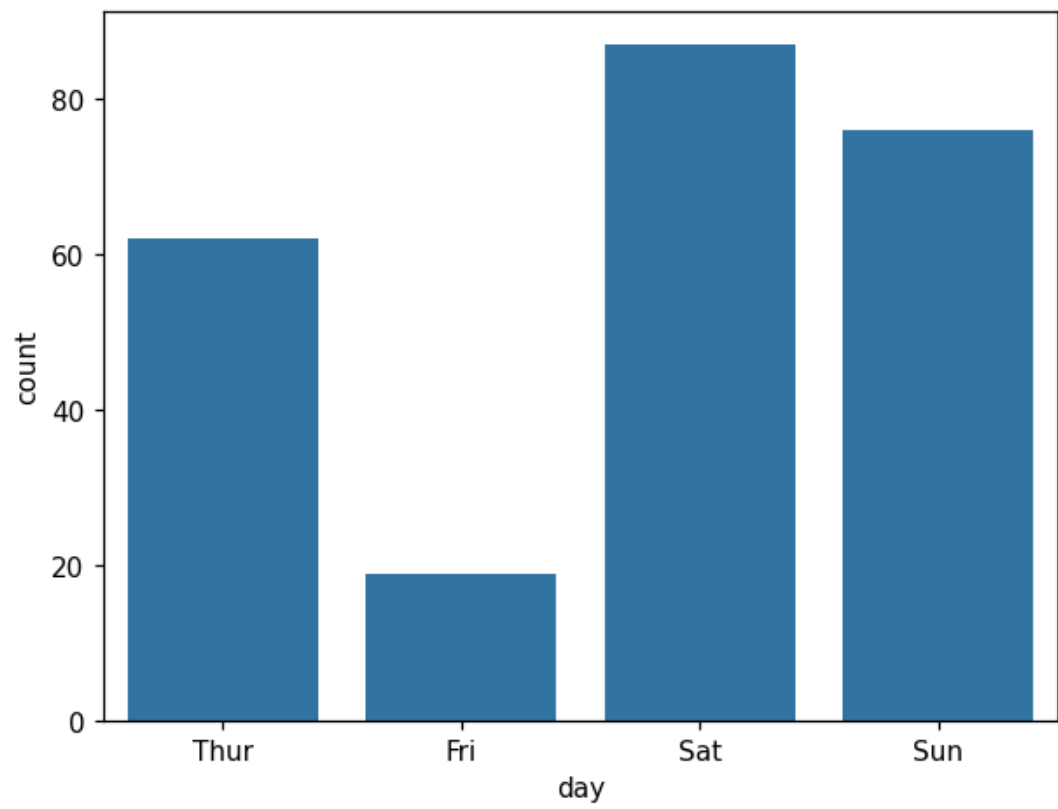


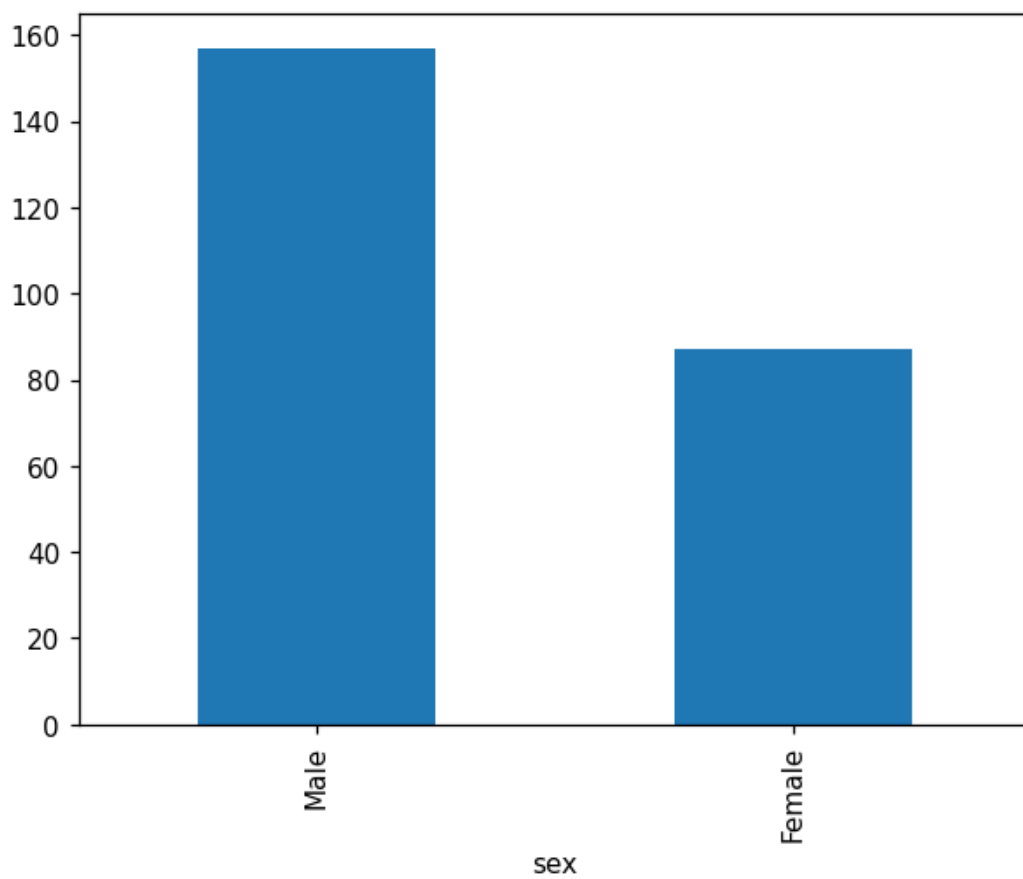
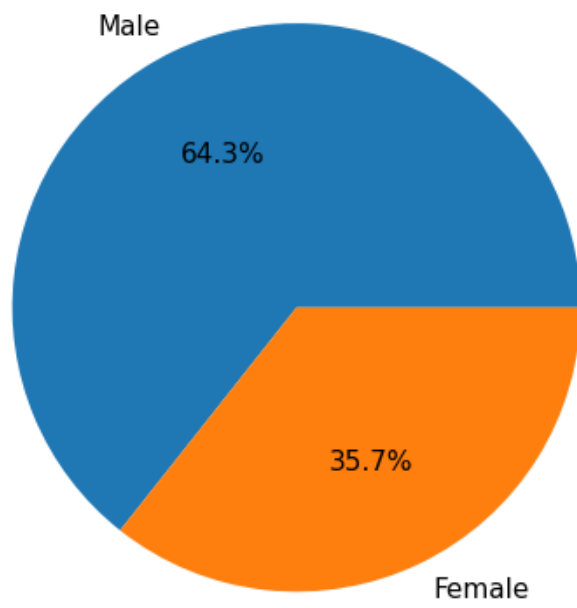


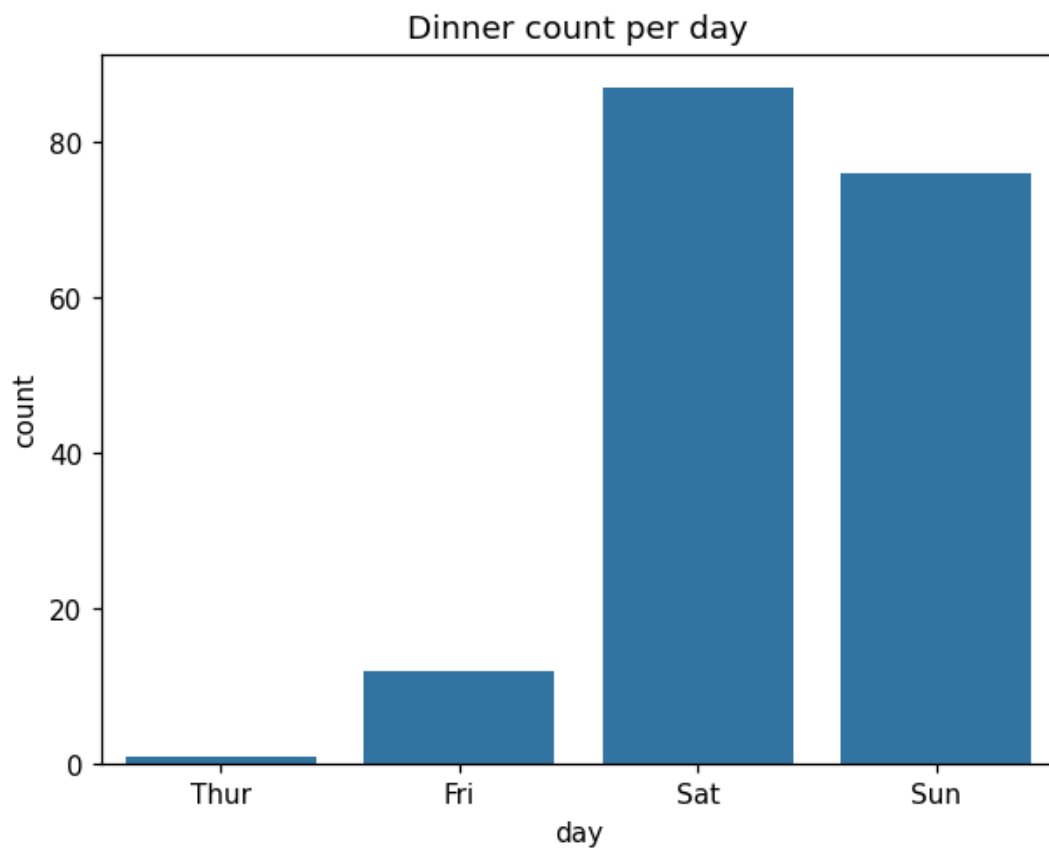
```

day
Sat      87
Sun      76
Thur     62
Fri      19
Name: count, dtype: int64
sex
Male     157
Female   87
Name: count, dtype: int64

```







RESULT

Thus exploratory data analysis (quantitative and qualitative) was performed on the tips data set. The total bill and tip are positively correlated, the bill amounts are right-skewed with a few high-value outliers, and most of the records belong to dinner time.